

트랜스크립트 파일을 이용한 비디오 게임의 자동화된 연령 등급 분류를 위한 텍스트 분석

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Textual Analysis for Automated Age Rating of Video Games From Transcript Files

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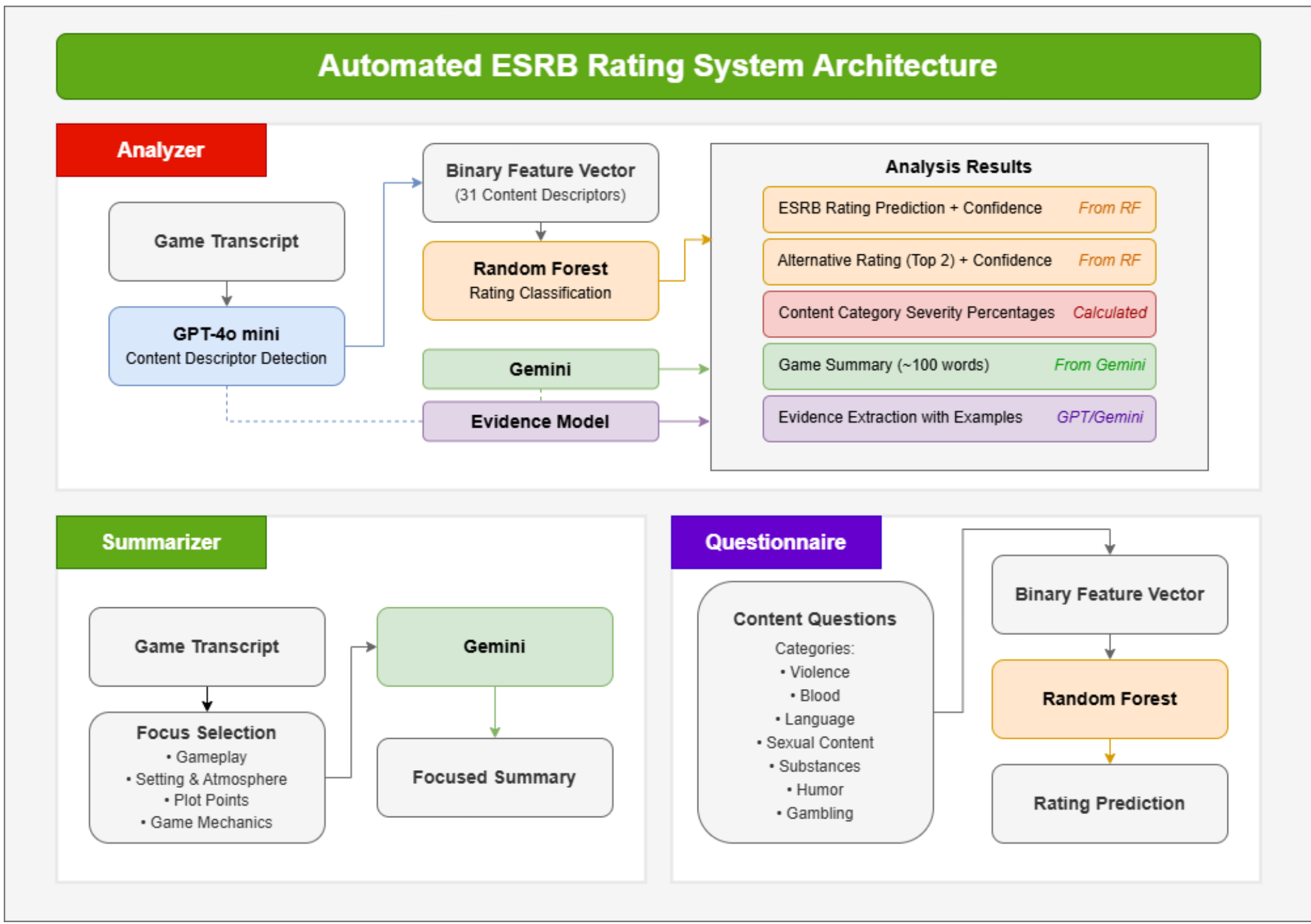
Overview

This research presents a two-stage pipeline approach for automated prediction of ESRB ratings from video game transcripts. The system combines GPT-4o mini's natural language understanding with a Random Forest classifier to analyze game text and predict appropriate age ratings with 81.7% accuracy

Methods

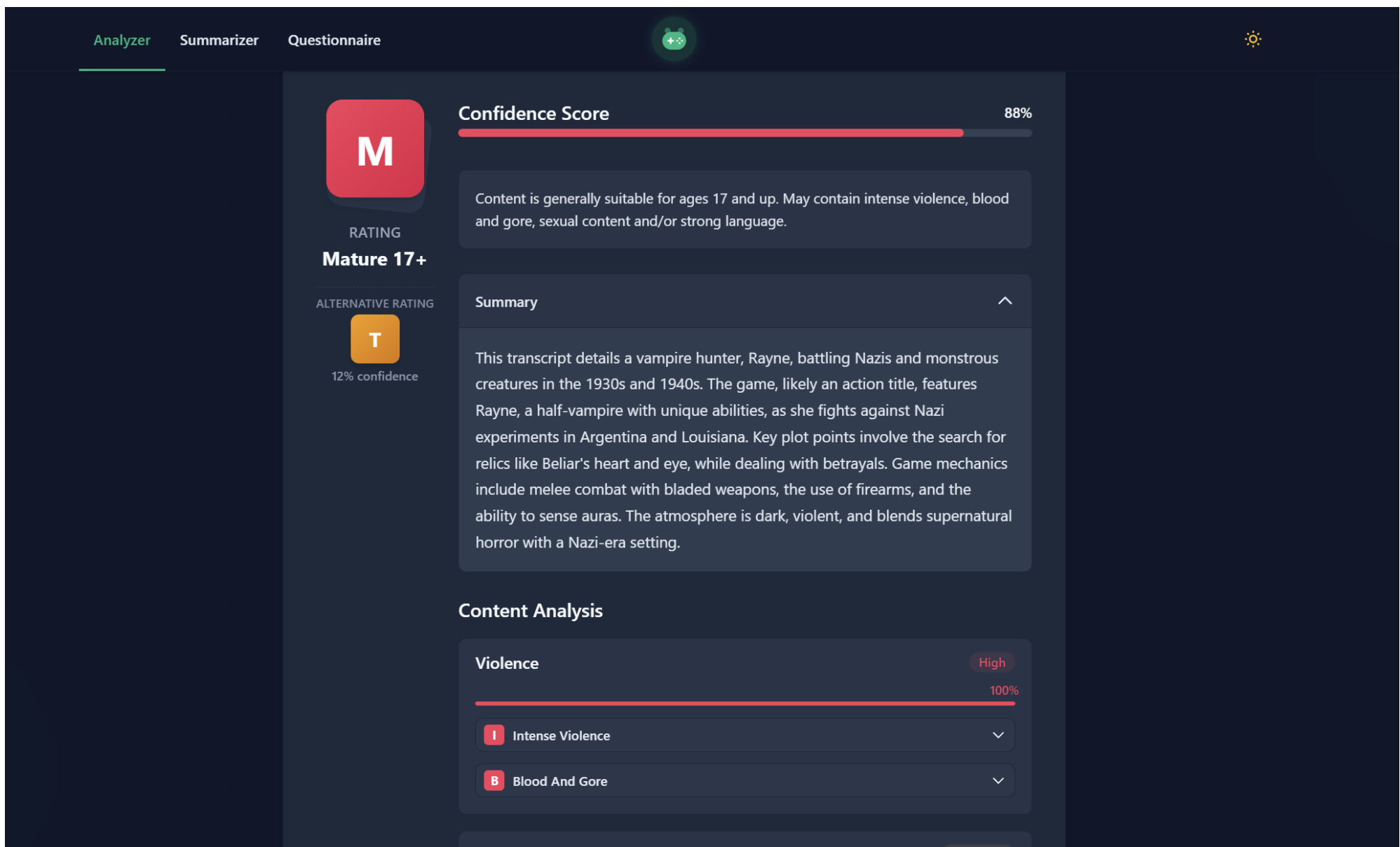
- GPT-4o mini analyzes game transcripts to identify 31 standardized ESRB content descriptors
- Random Forest classifier processes binary feature vectors to predict final rating (E, E10+, T, M)
- System extracts evidence and calculates severity percentages across content categories

System Configuration



Results

- 81.7% overall accuracy across 40 diverse game transcripts (1990–2024)
- Perfect prediction for Mature-rated games (100%)
- Strong performance for Everyone-rated titles (93.3%)
- Outperformed GPT-only approach (70.0%)
- Web application provides rating predictions with confidence scores, content severity analysis, summary, and evidence extraction



Conclusion

This automated system addresses critical industry challenges of resource intensity, cost, and consistency in content rating. It demonstrates significant potential for practical application in the gaming ecosystem as a tool for developers, parents, and potentially rating organizations.

Analyzer Page	Core component that processes game transcripts through GPT-4o-mini to detect content descriptors and uses Random Forest for rating classification
Summarizer Page	Uses Gemini to generate focused summaries (gameplay, plot, setting, mechanics) from game transcripts
Questionnaire Page	Alternative approach allowing direct input of content descriptors via questions about violence, language, etc.
Evidence Extraction	Identifies specific examples from transcript that support content descriptors and calculates severity percentages